

Lemma 1. *Let α be a fixed graph whose vertices are partitioned as*

$$V(\alpha) = U_\alpha \sqcup W_\alpha.$$

Let H_α be the bipartite graph with vertex classes U_α, W_α and edge set consisting only of the edges of α with one endpoint in U_α and one endpoint in W_α . Let

$$\tau(\alpha) = \min\{|S| : S \subseteq V(H_\alpha) \text{ is a vertex cover of } H_\alpha\}.$$

Then edges of α lying entirely inside U_α or entirely inside W_α do not change $\|M_\alpha\|$. Consequently it is enough to prove norm estimates for the bipartite cross-edge graph H_α .

Proof. For a realization φ , the product over same-side edges in U_α depends only on the row index $\varphi(U_\alpha)$. Thus those factors multiply rows by signs. Similarly, the product over same-side edges in W_α depends only on the column index and multiplies columns by signs. Multiplication on the left or right by a diagonal sign matrix is an isometry for the operator norm. Therefore the operator norm is unchanged after deleting all same-side edges.

For a bipartite graph, a set meeting every cross edge is exactly a vertex separator between the left and right sides in the sense that deleting it removes all left-right paths. Thus in this two-sided setting the minimum separator size is the minimum vertex-cover size $\tau(\alpha)$. $\square$

Lemma 2. *Let X and Y be finite sets with $|X|, |Y| \leq C_0 n$. Let $N_k(X, Y)$ be the number of closed alternating walks*

$$y_1, x_1, y_2, x_2, \dots, y_k, x_k, y_1, \quad x_i \in X, \quad y_i \in Y,$$

such that every edge of the complete bipartite graph $X \times Y$ traversed by the walk is traversed at least twice. Then

$$N_k(X, Y) \leq C^k n^{k+1},$$

whenever $k^{C_} \leq n$, where C and C_* depend only on C_0 . In particular this holds for $k = \lceil \log n \rceil$ and all sufficiently large n .*

Proof. For such a walk P , let $\Gamma(P)$ be its support graph, with q distinct vertices and e distinct edges. The support graph is connected, each of its edges is traversed at least twice, and the walk has length $2k$. Hence $e \leq k$ and $q \leq e + 1$. Put

$$\delta = k + 1 - q.$$

This is the vertex deficit from the tree-like case.

We need the following standard Füredi–Komlós encoding estimate. The number of canonical equality patterns of such walks with exactly $q = k + 1 - \delta$ support vertices is at most

$$C_1^k k^{C_1 \delta}. \tag{1}$$

Here “canonical” means that vertices on each side are named by their order of first appearance, so that a canonical pattern records only which positions of the walk have equal labels. For completeness we recall the encoding. Expose the walk from its root y_1 . First entrances into new vertices form a rooted plane tree. If $r = e - q + 1$ is the graph excess and $s = k - e$ is the multiplicity surplus, then

$$\delta = (k - e) + (e - q + 1) = s + r.$$

Mark as exceptional the $2s$ surplus traversals beyond the first two traversals of each support edge, and also the two mandatory traversals of each of the r non-tree support edges. After deleting

these exceptional traversals, the remaining walk traverses each edge of a rooted tree exactly twice, hence is the contour walk of a rooted plane tree; such contour data has at most 4^k possibilities. To reconstruct the original canonical walk, choose the exceptional times and, for each exceptional non-tree traversal, its endpoint among previously exposed vertex slots; this costs at most $k^{C_1(s+r)} = k^{C_1\delta}$. The remaining local choices are from a fixed finite alphabet of opening and closing moves and contribute only another C_1^k . This proves (1).

For any fixed canonical pattern with $q = k + 1 - \delta$ support vertices, the labels of the support vertices can be chosen in at most $(2C_0n)^q$ ways. Therefore

$$N_k(X, Y) \leq \sum_{\delta=0}^k C_2^k k^{C_2\delta} n^{k+1-\delta} = C_2^k n^{k+1} \sum_{\delta=0}^k \left(\frac{k^{C_2}}{n} \right)^\delta.$$

If $k^{C_*} \leq n$ with C_* large enough, the last sum is bounded by an absolute constant. Enlarging C proves the claimed estimate. $\square$

Lemma 3. *Let $G = (A, B, E)$ be a fixed bipartite graph with a matching saturating B . For each vertex $z \in A \cup B$, let Ω_z be a finite label set, and assume all nonempty Ω_z have sizes between cn and Cn , where $c, C > 0$ are fixed constants. The sets Ω_z are pairwise disjoint. Define*

$$Z_G(x, y) = \left(\prod_{a \in A} |\Omega_a|^{-1/2} \right) \prod_{(a,b) \in E} \varepsilon_{x_a, y_b},$$

with rows indexed by $x \in \prod_{a \in A} \Omega_a$ and columns indexed by $y \in \prod_{b \in B} \Omega_b$. Then

$$\mathbb{E} \|Z_G\|^2 \leq C_G.$$

Proof. It suffices to prove a high even moment bound. Let $a = |A|$, $b = |B|$, and choose a matching $b' \mapsto a_{b'}$ saturating B . For $k \geq 1$, expand

$$\mathbb{E} \operatorname{tr}((Z_G^\top Z_G)^k).$$

The trace expansion sums over k row labelings $x^1, \dots, x^k$ and k column labelings $y^1, \dots, y^k$, with $y^{k+1} = y^1$. The normalization contributes $\prod_{a \in A} |\Omega_a|^{-k}$.

If a summand is nonzero, each Rademacher variable appears an even number of times. Since the sets Ω_z are pairwise disjoint, a Rademacher variable contributed by the shape edge $(a_{b'}, b')$ cannot coincide with a variable contributed by any different shape edge. Thus the parity constraint for this matched edge must hold inside the factors coming from this edge alone. For $(a_{b'}, b')$, the labels

$$y_{b'}^1, x_{a_{b'}}^1, y_{b'}^2, x_{a_{b'}}^2, \dots, y_{b'}^k, x_{a_{b'}}^k, y_{b'}^1$$

form a closed alternating walk in the complete bipartite graph $\Omega_{b'} \times \Omega_{a_{b'}}$, and every traversed edge is repeated. Lemma 2 gives at most $C_G^k n^{k+1}$ choices for these $2k$ labels whenever $k^{C_G} \leq n$.

Doing this for all $b' \in B$ gives at most $C_G^k n^{b(k+1)}$ choices for matched coordinates. The remaining $a - b$ row coordinates contribute at most $C_G^k n^{k(a-b)}$ choices. Extra edges of G only add parity constraints. Since all $|\Omega_a|$ are comparable to n ,

$$\mathbb{E} \operatorname{tr}((Z_G^\top Z_G)^k) \leq C_G^k n^{-ak} n^{b(k+1)} n^{k(a-b)} = C_G^k n^b.$$

Taking $k = \lceil \log n \rceil$, which satisfies $k^{C_G} \leq n$ for all sufficiently large n , we get

$$\mathbb{E} \|Z_G\|^2 \leq \left(\mathbb{E} \|Z_G\|^{2k} \right)^{1/k} \leq \left(C_G^k n^b \right)^{1/k} \leq C'_G.$$

$\square$

Lemma 4. Let $H = (U, W, E)$ be a fixed bipartite graph. Let τ be its minimum vertex-cover size. For each $z \in U \cup W$, let $\Omega_z \subset [n]$ be pairwise disjoint label sets, all of sizes between cn and Cn . Let $M_{H,\Omega}$ be the matrix with rows indexed by $\prod_{u \in U} \Omega_u$, columns indexed by $\prod_{w \in W} \Omega_w$, and entries

$$M_{H,\Omega}(x, y) = \prod_{(u,w) \in E} \varepsilon_{x_u, y_w}.$$

Then

$$\mathbb{E} \|M_{H,\Omega}\| \leq C_H n^{(|U|+|W|-\tau)/2}.$$

Proof. Let S be a minimum vertex cover of H , and write

$$S_U = S \cap U, \quad S_W = S \cap W, \quad I_U = U \setminus S, \quad I_W = W \setminus S.$$

By König's theorem, H has a matching of size $\tau = |S|$, and every such maximum matching saturates S . Hence every vertex of S_W is matched to a distinct vertex of I_U , and every vertex of S_U is matched to a distinct vertex of I_W .

Define normalized feature matrices

$$Z_L(x, r) = \left(\prod_{u \in I_U} |\Omega_u|^{-1/2} \right) \prod_{(u,w) \in E, u \in I_U, w \in S_W} \varepsilon_{x_u, r_w},$$

with rows $x \in \prod_{u \in I_U} \Omega_u$ and columns $r \in \prod_{w \in S_W} \Omega_w$, and

$$Z_R(y, \ell) = \left(\prod_{w \in I_W} |\Omega_w|^{-1/2} \right) \prod_{(u,w) \in E, u \in S_U, w \in I_W} \varepsilon_{\ell_u, y_w}.$$

The matching observation is exactly the hypothesis of Lemma 3 for both Z_L and Z_R , so $\mathbb{E} \|Z_L\|^2$ and $\mathbb{E} \|Z_R\|^2$ are bounded by constants depending only on H .

Because S covers all edges, there are no edges between I_U and I_W . For unit vectors ξ, η , group ξ by $\ell \in \prod_{S_U} \Omega$ and η by $r \in \prod_{S_W} \Omega$. The bilinear form factors as

$$|\langle \xi, M_{H,\Omega} \eta \rangle| \leq \left(\prod_{z \in I_U \cup I_W} |\Omega_z|^{1/2} \right) \left(\sum_{\ell, r} |\langle \xi_\ell, Z_L(\cdot, r) \rangle|^2 \right)^{1/2} \left(\sum_{\ell, r} |\langle \eta_r, Z_R(\cdot, \ell) \rangle|^2 \right)^{1/2}.$$

The two sums are at most $\|Z_L\|^2$ and $\|Z_R\|^2$, respectively. Hence

$$\|M_{H,\Omega}\| \leq \left(\prod_{z \in I_U \cup I_W} |\Omega_z|^{1/2} \right) \|Z_L\| \|Z_R\|.$$

Taking expectation and using Cauchy-Schwarz gives

$$\mathbb{E} \|M_{H,\Omega}\| \leq C_H n^{(|I_U|+|I_W|)/2} = C_H n^{(|U|+|W|-\tau)/2}.$$

□

Lemma 5. For the injective graph matrix M_α in the problem statement,

$$\mathbb{E} \|M_\alpha\| \leq C_\alpha n^{(|V(\alpha)|-\tau(\alpha))/2}.$$

Proof. By Lemma 1, delete same-side edges and let H_α be the remaining bipartite graph. Put $v = |V(\alpha)|$. Choose uniformly a balanced ordered partition

$$[n] = \bigsqcup_{a \in V(\alpha)} \Omega_a$$

with all $|\Omega_a|$ equal to either $\lfloor n/v \rfloor$ or $\lceil n/v \rceil$.

For such a partition Ω , let M^Ω be the matrix obtained by retaining only those realization terms for which the label of vertex a lies in Ω_a . Since the color classes are disjoint, this is exactly a separated-label matrix of the form in Lemma 4, embedded into the original row and column spaces.

For any fixed injective realization φ , the probability over the random balanced partition that $\varphi(a) \in \Omega_a$ for every $a \in V(\alpha)$ is

$$p_n = \frac{\prod_{a \in V(\alpha)} |\Omega_a|}{n(n-1) \cdots (n-v+1)},$$

which is bounded below by a positive constant depending only on α . Therefore, entrywise,

$$\mathbb{E}_\Omega M^\Omega = p_n M_\alpha.$$

Thus

$$\|M_\alpha\| \leq p_n^{-1} \mathbb{E}_\Omega \|M^\Omega\|.$$

Taking expectation over the Rademacher signs and applying Lemma 4 to each balanced partition gives

$$\mathbb{E} \|M_\alpha\| \leq C_\alpha n^{(|V(\alpha)| - \tau(\alpha))/2}.$$

□

Lemma 6. *For the injective graph matrix M_α ,*

$$\|M_\alpha\| \geq c_\alpha n^{(|V(\alpha)| - \tau(\alpha))/2}$$

for all sufficiently large n . Consequently the same lower bound holds for $\mathbb{E} \|M_\alpha\|$.

Proof. Again delete same-side edges, which only apply row and column sign changes. Let S be a minimum vertex cover of the cross-edge graph, and write

$$S_U = S \cap U_\alpha, \quad S_W = S \cap W_\alpha, \quad I_U = U_\alpha \setminus S, \quad I_W = W_\alpha \setminus S.$$

Fix one injective assignment of distinct labels to the vertices of S . Then restrict further to rows and columns whose free coordinates lie in the complement of these fixed labels, are internally distinct, and are disjoint from each other. In other words, after the S -labels have been fixed, the remaining row coordinates range over ordered distinct I_U -tuples from an $N = n - |S|$ point set, the remaining column coordinates range over ordered distinct I_W -tuples from the same set, and a matrix entry is retained exactly when the two free tuples are disjoint.

Since $I_U \cup I_W$ is independent in the cross-edge graph, no remaining edge joins a free I_U -vertex to a free I_W -vertex. Therefore, on this restricted submatrix, all random signs factor as a row sign times a column sign times one fixed sign. Thus the restricted submatrix is obtained from the deterministic disjointness matrix

$$D_{p,q,N}, \quad p = |I_U|, \quad q = |I_W|, \quad N = n - |S|,$$

by multiplying rows and columns by signs. Here $D_{p,q,N}$ has rows indexed by ordered distinct p -tuples from an N -point set, columns indexed by ordered distinct q -tuples from the same set, and entry 1 exactly when the two tuples are disjoint.

The norm of $D_{p,q,N}$ is $\Theta_{p,q}(N^{(p+q)/2})$. The upper bound is its Frobenius norm. For the lower bound, apply it to the all-ones vector on columns: every row has

$$(N-p)(N-p-1)\cdots(N-p-q+1) = \Theta_{p,q}(N^q)$$

ones, and the number of rows is $\Theta_p(N^p)$ while the number of columns is $\Theta_q(N^q)$. Hence the all-ones test vector gives a singular value $\Theta_{p,q}(N^{(p+q)/2})$.

The restricted submatrix is a submatrix of M_α , so

$$\|M_\alpha\| \geq c_\alpha n^{(p+q)/2} = c_\alpha n^{(|V(\alpha)|-\tau(\alpha))/2}.$$

□

Theorem 1. *Let $(\varepsilon_{i,j})_{i < j}$ are i.i.d. Rademacher random variables (i.e., uniform in $\{-1, +1\}$). In addition, let α be a shape (a fixed graph) in which we partition the vertices $V(\alpha)$ into the left (U_α) and right (W_α) sides, and let n be a large integer. A realization is any injective map $\varphi : V(\alpha) \rightarrow [n] = \{1, \dots, n\}$ from the shape vertices to the ground set $[n]$. The graph matrix M_α is the $n^{|U_\alpha|} \times n^{|W_\alpha|}$ random matrix, whose rows and columns are indexed by ordered subsets of $[n]$ with cardinalities $|U_\alpha|$ and $|W_\alpha|$, respectively, given by*

$$M_\alpha = \sum_{\text{realization } \varphi} \left(\prod_{(i,j) \in E(\alpha)} \varepsilon_{\varphi(i), \varphi(j)} \right) e_{\varphi(U_\alpha)} e_{\varphi(W_\alpha)}^\top.$$

Prove or disprove that there are functions f and g depending only on the shape α and a constant $C = C(\alpha)$, such that

$$C^{-1} n^{f(\alpha)} (\log n)^{g(\alpha)} \leq \mathbb{E} [\|M_\alpha\|] \leq C n^{f(\alpha)} (\log n)^{g(\alpha)}.$$

Proof. The assertion is true. Let H_α be the bipartite graph of cross edges of α , and let

$$\tau(\alpha) = \min\{|S| : S \subseteq V(H_\alpha) \text{ meets every edge of } H_\alpha\}.$$

Define

$$f(\alpha) = \frac{|V(\alpha)| - \tau(\alpha)}{2}, \quad g(\alpha) = 0.$$

Lemma 5 gives

$$\mathbb{E} \|M_\alpha\| \leq C_\alpha n^{(|V(\alpha)|-\tau(\alpha))/2}.$$

Lemma 6 gives

$$\mathbb{E} \|M_\alpha\| \geq c_\alpha n^{(|V(\alpha)|-\tau(\alpha))/2}.$$

Absorbing c_α^{-1} and C_α into one constant $C = C(\alpha)$, we obtain

$$C^{-1} n^{f(\alpha)} (\log n)^{g(\alpha)} \leq \mathbb{E} \|M_\alpha\| \leq C n^{f(\alpha)} (\log n)^{g(\alpha)}.$$

Thus the required functions exist, and in fact no logarithmic factor is needed in this two-sided no-hidden-vertices formulation. □